Figure 8.1 A four-node clustered `ActorSystem`

The cluster module's ultimate goal is to provide fully automated features for actor distribution and failover. Let's concentrate on the following features:

- *Cluster membership*—Fault-tolerant membership for `ActorSystem`s.
- *Partition points*—An `ActorSystem` can be partitioned into actor subtrees located on different nodes. We discuss this and roles later in the chapter.

A single-purpose data-processing application is a good example of a candidate application for using clusters: for example, data-processing tasks like image recognition and real-time analysis of social media. Nodes can be added or removed when more or less processing power is required. Processing jobs are supervised: if an actor fails, the job can be restarted and retried on the cluster until it succeeds or is abandoned.

Figure 8.2 shows an example, but don't worry about the details (once you learn the basic concepts of clustering, we'll look at an example in depth). The goal is to develop an app that counts words: how many occurrences of a word are in a series of texts.